

Assignment 1:

Now that you've reached this section of the program, it's time to start thinking about your capstone project. Your first task is to draft a problem statement, which should include the following:

- A general overview of the question you will be asking/solving for (1-2 sentences).
- The data you think you'll need to answer this question (no official data is needed at this time, just ideas of what type of data will help you).
- List 1–3 techniques you might use to answer this question based on what you've learned so far.

The average completion time for this assignment is 120 minutes.

Required Capstone Assignment 11.1: Draft the Problem Statement

Required Capstone Assignment 11.1: Draft the Problem Statement

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeAn overview of the question to be solved in the capstone.	5 pts Full Marks	0 pts No Marks	5 pts
This criterion is linked to a Learning OutcomeIdentification of the type of data that will be used to solve the question.	5 pts Full Marks	0 pts No Marks	5 pts
This criterion is linked to a Learning OutcomeList of 1–3 techniques that could be used to answer the question identified.	5 pts Full Marks	0 pts No Marks	5 pts
Total Points: 15			

Assignment 2:

By this point in the program, you will have posed a question to answer in your final capstone submission and met with your Learning Facilitator to refine your question and generate ideas for datasets and techniques. Now it is time to submit the official proposal or problem statement.

On this discussion board, your initial post should contain the following:

- The research question you intend to answer (in one sentence, if possible)
- Your expected data source(s) (as either a link to existing data or a sentence describing where you will source the data from)
- The techniques you expect to use in your analysis
- The expected results
- Why this question is important

That last bullet point is the most important. Remember that your final capstone project will be graded not only on the execution of your plan but also on your ability to translate those results into actionable business intelligence that can be acted on by non-AI/ML personnel. Not only will you have to do the number crunching, but you will also have to distill the results into something that the general public can understand. This last bullet point is your opportunity to explain (in non-AI/ML language) why this question is important. What happens if this question is unanswered? What benefit will your analysis bring?

Assignment 3:

Overview

In this module, you will work on performing exploratory data analysis (EDA) to develop an initial report for your capstone project. You will use EDA to see what data can reveal beyond the formal modeling, hypothesis testing task, and data training to provide a better understanding of dataset variables and the relationships between them. You are encouraged to spend your time in this module cleaning your data and use feature engineering and EDA techniques to create visualizations to make sense of your findings. Additionally, you will also be required to use one of the ML algorithms you have learned so far in the program to develop a baseline model to use as a comparison in Module 24. You will have time in Module 24 to include additional models, clean the code, and make your work presentable for technical and non-technical audiences. For now, you will do the 'heavy lifting' of finding the answer to your research question.

The only deliverables for this module will be the Jupyter notebook(s) where you did your analysis and the updated [README.md Download README.md](#) file with the results section completed. If you have changed methodologies or datasets since Module 16, please make sure you reflect those changes in the [README.md Download README.md](#) file as well.

Submission Instructions:

- Post both your [README.md Download README.md](#) file and your Jupyter notebooks to your GitHub Repository. Submit a link to your repository here.

The average completion time for this assignment is 15 hours. **Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)**

Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeProject organization: • README with a summary of findings and a link to the Jupyter notebook. • Jupyter notebook has headings and text appropriately formatted. • No unnecessary files. • Directories and files have appropriate names and locations.	5 pts Excellent Your submission includes all of the listed components.	0 pts Criterion Not Met Your submission includes a f of the listed components.	5 pt
			5 pt

*The average completion time for this assignment is 15 hours.***Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)**

Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeSyntax and code quality: - Libraries are imported and aliased correctly. - Code does not contain errors. - No long strings of code output. - Demonstrates competency with pandas. - Demonstrates competency with visualization libraries e.g., seaborn, Matplotlib, and Plotly. - Comments are used appropriately to explain code. - Variables are sensible.	5 pts Excellent Your submission includes all of the listed components.	0 pts Criterion Not Met Your submission includes a f	5 pt
			5 pt

*The average completion time for this assignment is 15 hours.***Required Capstone Assignment**
20.1: Initial Report and Exploratory Data Analysis (EDA)

Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeVisualizations: • Appropriate plots for categorical and continuous variables are utilized. • Plots contain readable labels. • Plots contain descriptive titles. • Axes are legible. • Subplots are used when appropriate. • Plots are scaled appropriately for readability.	5 pts Excellent Your submission includes all of the listed components.	0 pts Criterion Not Met Your submission includes a few of the listed components.	5 pts

*The average completion time for this assignment is 15 hours.***Required Capstone Assignment**
20.1: Initial Report and Exploratory Data Analysis (EDA)

Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeData Cleaning and EDA • The dataset is clean: imputation and removal of missing values, removal of duplicate values • Outliers analysis to identify anomalies in the dataset. • Perform feature engineering to extraction and transformation of variables from raw data.	5 pts Excellent Your submission includes all of the listed components.	0 pts Criterion Not Met Your submission includes a f	5 pt s none

*The average completion time for this assignment is 15 hours.***Required Capstone Assignment**
20.1: Initial Report and Exploratory Data Analysis (EDA)

Required Capstone Assignment 20.1: Initial Report and Exploratory Data Analysis (EDA)

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeModeling : • Develop an appropriate classification or regression ML model to utilize as baseline for your analysis. • Valid interpretation of an evaluation metric. • Clear identification of an evaluation metric. • Clear rationale for use of the given evaluation metric.			5 pts
	5 pts Excellent Your submission includes all of the listed components.	0 pts Criterion Not Met Your submission includes a f of the listed components.	
Total Points: 25			

Final Capstone Project:

Overview

In Module 16, you submitted your Jupyter Notebooks and the results section of your README file. Here, you will submit your final technical findings and the nontechnical report of your capstone:

- The technical component should be delivered in the form of a collection of Jupyter Notebooks that walk through your analysis.
- The nontechnical report should be delivered in the form of a README that covers your problem, results, important findings, and suggestions for next steps.

Make sure to polish your presentation of the findings:

- Look through all your code to ensure there are no errors.
- Provide meaningful headers and comments in your Jupyter Notebooks so those who look through your work can clearly understand your reasoning.
- Ensure your [README.md](#) [Download README.md](#) file has a clear summary of your project and is written in language that is readable for those who do not have a background in ML/AI.
- For more details as to what should be included in your README file, please see the following [template](#) [Download template](#).

Use the provided rubric for a full list of project requirements.

Exemplars

You are encouraged to look through the two example capstones linked below to help you get a sense of what an exemplary completed capstone looks like. These examples are being shared with consent of the authors.

- [Detecting Suicide Ideation](#)[Links to an external site.](#)
- [Predictive Maintenance](#)[Links to an external site.](#)

Submission Instructions:

- Submit the link to your GitHub repository here. Your Jupyter Notebooks and README should be easily accessible there.

The average completion time for this assignment is 900 minutes.

Capstone

Capstone

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeProject Organization: · README file with a summary of findings and link to your Jupyter Notebook · Jupyter Notebook with headings and text appropriately formatted · No unnecessary files · Directories and files with appropriate names and location	10 pts Excellent Your submission includes all of the listed components.	0 pts Criterion not met Your submission includes a few none of the listed components	10 pts

Capstone

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeSyntax and Code Quality: · Libraries imported and named correctly · Code with no errors · No long strings of code output · Demonstration of competency with pandas · Demonstration of competency with visualization libraries, e.g., seaborn, Matplotlib, and Plotly · Appropriate use of comments to explain code · Sensible variables	10 pts Excellent Your submission includes all of the listed components.	0 pts Criterion not met Your submission includes a few none of the listed components	10 pts

Capstone

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeVisualizations: · Appropriate plots for categorical and continuous variables · Plots with human-readable labels · Plots with descriptive titles · Legible axes · Appropriate use of subplots when necessary · Plots that are scaled appropriately for readability	10 pts Excellent Your submission includes all of the listed components.	0 pts Criterion not met Your submission includes a few none of the listed components	10 pts

Capstone

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeModeling: · Multiple regression or classification models · Cross-validation of models · Grid Search hyperparameters · Appropriate interpretation of models · Appropriate interpretation of evaluation metric · Clear identification of evaluation metric · Clear rationale for use of given evaluation metric			10 pts
	10 pts Excellent Your submission includes all of the listed components.	0 pts Criterion not met Your submission includes a few none of the listed componen	

Capstone

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeFindings: · Clearly states business understanding of the problem · Clean and organized notebook with data cleaning · Correct and concise interpretation of descriptive and inferential statistics · Clearly states findings in their own sections, with actionable items highlighted in appropriate language for a nontechnical audience · Next steps and recommendations			10 pts
	10 pts Excellent Your submission includes all of the listed components.	0 pts Criterion not met Your submission includes a few none of the listed components	
Total Points: 50			